

Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 7 Extension

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A. Which equation? $a = \frac{v-u}{t}$ or $v^2 - u^2 = 2as$ — the empty line in your Givens box tells you.

- 1) **Calculate** the speed a go-kart was moving at before it accelerated at 2.5 m/s^2 for 4.0 s , reaching 18 m/s . [3]

Givens $s =$ $u = ?$ $v = 18 \text{ m/s}$ $a = 2.5 \text{ m/s}^2$ $t = 4.0 \text{ s}$

No s , so $a = \frac{v-u}{t}$

$$2.5 = \frac{18-u}{4.0} \Rightarrow 18-u=10 \Rightarrow u = \mathbf{8.0 \text{ m/s}}$$

- 2) **Calculate** the acceleration of a motorbike that brakes from 24 m/s to a stop in 48 m . [3]

Givens $s = 48 \text{ m}$ $u = 24 \text{ m/s}$ $v = 0$ $a = ?$ $t =$

No t , so $v^2 - u^2 = 2as$

$$v^2 - u^2 = 2as$$

$$0^2 - 24^2 = 2 \times a \times 48$$

$$a = \frac{0^2 - 24^2}{2 \times 48} = \frac{-576}{96} = \mathbf{-6.0 \text{ m/s}^2}$$

- 3) **Calculate** the speed of a hockey puck after it slides 40 m across the ice. It is hit at 5.0 m/s and slows down with an acceleration of -0.20 m/s^2 . [3]

Givens $s = 40 \text{ m}$ $u = 5.0 \text{ m/s}$ $v = ?$ $a = -0.20 \text{ m/s}^2$ $t =$

No t , so $v^2 - u^2 = 2as$

$$v^2 - u^2 = 2as$$

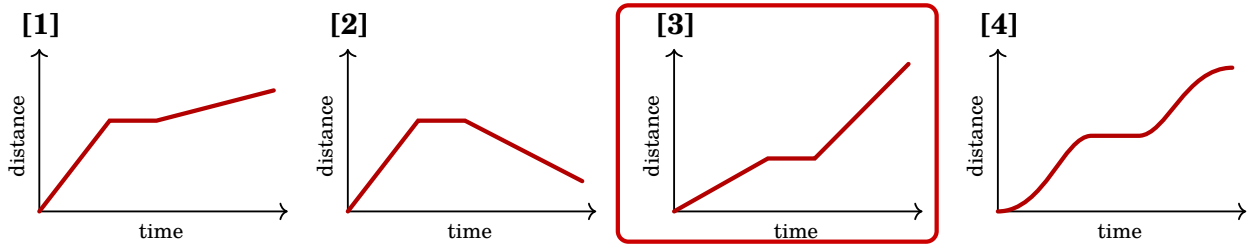
$$v^2 - 5.0^2 = 2 \times (-0.20) \times 40$$

$$v^2 = 5.0^2 + 2 \times (-0.20) \times 40 = 25 - 16 = 9.0$$

$$v = \sqrt{9.0} = \mathbf{3.0 \text{ m/s}}$$

B. Graphs choose the graph that matches.

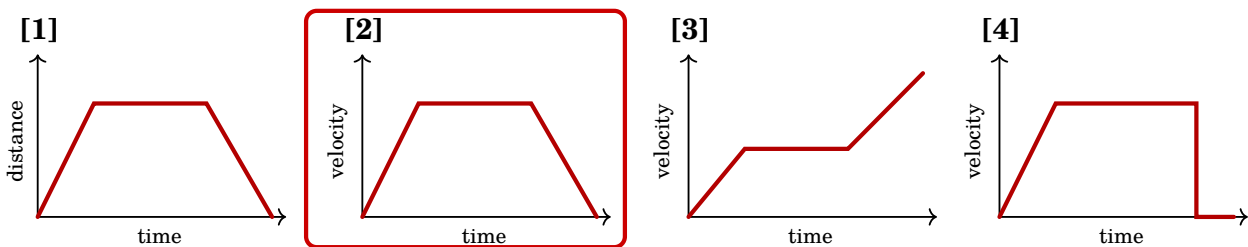
- 4) A bus drives at a steady speed, stops at a bus stop, then drives on **faster**. Which distance–time graph shows this journey? [2]



Graph [3] because

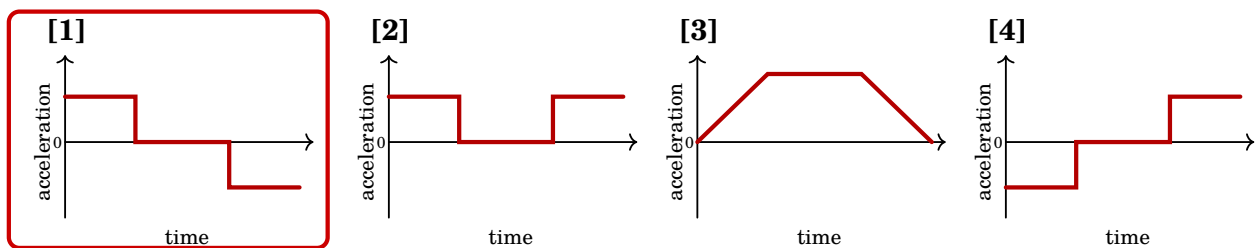
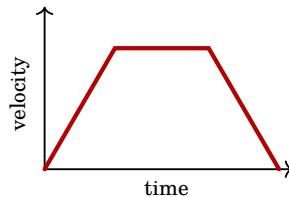
the line is flat (stationary) in the middle, and steeper after the stop than before it (faster).

- 5) A lift speeds up steadily from rest, moves at a constant speed, then slows steadily to a stop. Which graph shows this? [1]



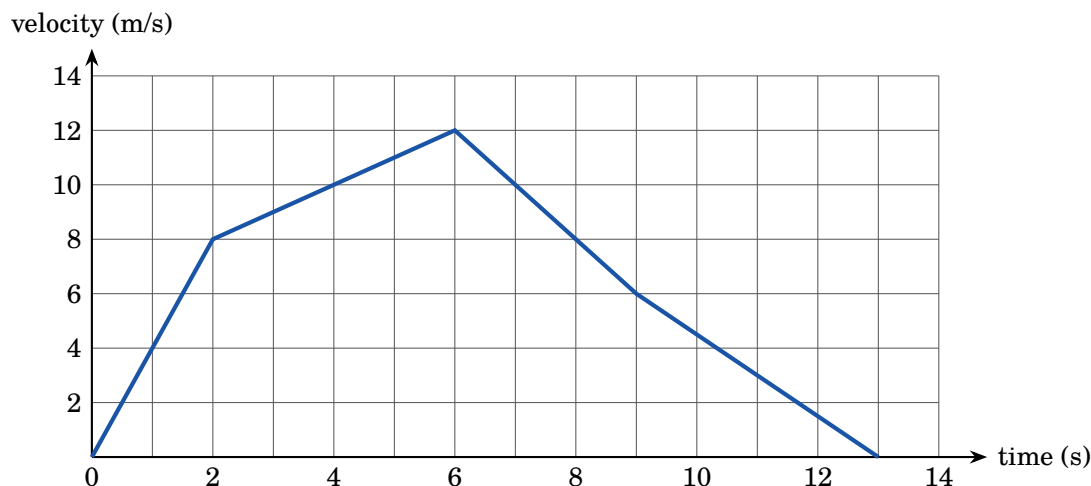
Graph [2] — [1] is a distance axis; [4] stops in no time at all.

- 6) Here is a velocity–time graph. Which acceleration–time graph matches it? [1]



Graph [1] — gradient positive, then zero, then negative.

7) A motor boat leaves a jetty, then slows down to moor. The graph shows its motion.



(a) Between which times is the acceleration greatest? Explain how the graph shows this.

0 to 2 s — the line is steepest there (greatest gradient: 4.0 m/s^2).

[2]

(b) Calculate the acceleration between 2 s and 6 s.

$$a = \frac{12 - 8}{6 - 2} = \frac{4}{4} = 1.0 \text{ m/s}^2$$

[1]

(c) Describe the motion of the boat from 6 s to 13 s.

It slows down from 12 m/s to 6 m/s (deceleration 2.0 m/s^2), then slows more gently (1.5 m/s^2) until it stops at 13 s.

[2]

(d) Use $v^2 - u^2 = 2as$ and your answer to (b) to calculate the distance travelled between 2 s and 6 s.

Givens

$$s = ?$$

$$u = 8 \text{ m/s}$$

$$v = 12 \text{ m/s}$$

$$a = 1.0 \text{ m/s}^2$$

$$t = 4 \text{ s}$$

$$v^2 - u^2 = 2as$$

$$12^2 - 8^2 = 2 \times 1.0 \times s$$

$$s = \frac{12^2 - 8^2}{2 \times 1.0} = \frac{80}{2.0} = 40 \text{ m}$$

[2]

(e) Use the area under the graph to calculate the total distance the boat travels. The area between 2 s and 6 s matches your answer to (d).

$$\begin{aligned} & \frac{1}{2} \times 2 \times 8 + \frac{1}{2} \times (8 + 12) \times 4 + \frac{1}{2} \times (12 + 6) \times 3 + \frac{1}{2} \times 4 \times 6 \\ &= 8 + 40 + 27 + 12 = 87 \text{ m} \text{ — the 2–6 s trapezium is } 40 \text{ m, the same as (d).} \end{aligned}$$

[3]